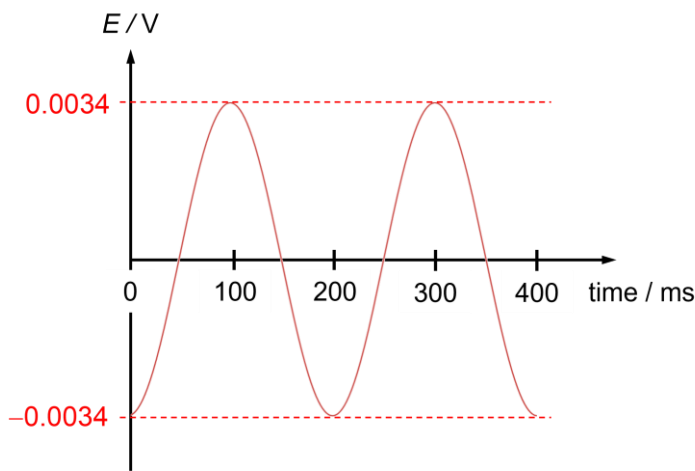


Qn	Suggested Solutions
5(a)	Faraday's law states that the magnitude of the induced e.m.f. in a conductor is directly proportional to the rate of change of magnetic flux linkage experienced by the conductor.
(b)	Magnetic flux density in the solenoid is constant. Since the cross sectional area and number of turns of coil C remains the same, the magnetic flux linkage is constant.
(c)(i)	$\omega = \frac{2\pi}{T}$ $T = \frac{2\pi}{\omega}$ $= \frac{2\pi}{10\pi}$
	$T = 0.20 \text{ s}$ $= 200 \text{ ms (Shown)}$
(ii)	$B_{\max} = \mu_0 n I_{\max}$ $= (4\pi \times 10^{-7})(4000)(4.8)$ $= 0.024127 \text{ T}$
	$\Phi_{\max} = N B_{\max} A \cos 0^\circ$ $= (71)(0.024127)(0.64 \times 10^{-4})$
	$\Phi_{\max} = 1.096 \times 10^{-4} \text{ Wb}$ $= 1.1 \times 10^{-4} \text{ Wb (Shown)}$
(iii)	$\Phi = N B A \sin(\omega t)$ $= N A (\mu_0 n I_0) \sin(\omega t)$ $= \Phi_0 \sin(\omega t)$ $\varepsilon = -\frac{d\Phi}{dt} = -\omega \Phi_0 \cos(\omega t)$
	$\varepsilon_{\max} = \omega \Phi_0$ $= (10\pi)(1.1 \times 10^{-4})$
	$\varepsilon_{\max} = 0.0034 \text{ V}$
(iv)	

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